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Quality assurance mechanisms and the sustainability of globalized higher education systems: a systems theory-based mixed-methods study

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Introduction: As the world becomes more interconnected, educational institutions must contend with the difficulty of upholding high standards in various cultural and regulatory landscapes. Effective quality assurance techniques that address differences in education practices and expectations are crucial for ensuring excellence in international educational systems. This study examines the impact of quality assurance mechanisms, particularly capacity, capability, and self-evaluation, on the sustainability of globalized education systems. It aims to determine whether these approaches consistently support high standards across diverse educational systems.

Methods: Employing a mixed-methods design, the research integrates data from a structured questionnaire ($n = 400$), in-depth interviews with students, lecturers, and QA officers ($n = 15$), and an analysis of institutional policy documents. This triangulated approach enhances validity by capturing both quantitative trends and qualitative depth.

Results: The findings show that the sustainability of globalized education systems is significantly influenced by effective quality assurance management.

Discussion: This study emphasizes globalization and acknowledges the growing interconnection of educational institutions, as well as the need for quality assurance. The study contributes a theoretically grounded systems-theoretic model, offering strategic insights for context-aware QA frameworks across diverse global settings.

KEYWORDS

capacity and capability, globalization, quality assurance, self-evaluation, sustainability

Introduction

Quality assurance (QA) plays a crucial role in ensuring consistent excellence across diverse cultural and regulatory landscapes, which is vital for the sustainability of global educational systems. To maintain academic standards and encourage long-term growth in our increasingly interconnected world, effective quality assurance mechanisms, like

self-evaluation, professional development, and institutional capacity, are key. When there are gaps in policy and practice, it can hinder cohesion and quality in international education, making this issue especially relevant. Research conducted at public universities in Nigeria shows that well-structured quality assurance processes significantly enhance educational sustainability, underscoring the importance of data-driven assessments and continuous improvement. The findings suggest that we need standardized QA frameworks that can adapt to regional needs while advancing the broader goals of global education, ensuring that quality is measurable and transferable across borders.

While numerous studies have explored quality assurance in education, [Mogbeyiteren et al. \(2019\)](#) specifically examined its impact on school administration in Nigerian institutions. Their paper argued that to oversee education in Nigerian schools and meet societal demands effectively, we must address challenges like overcrowding, poor facilities, insufficient staffing, lack of supervision, and outdated research, all of which contribute to the country's declining educational standards. High-quality education is, therefore, the foundation of sustainable development. The descriptive analysis revealed that the quality of education for Nigerians has suffered due to inadequate quality assurance in schools; thus, the study relied solely on qualitative data from secondary sources. As a result, it recommended a healthy implementation of quality assurance practices. [Anene \(2021\)](#) explored the crucial role of quality assurance in achieving sustainable quality education within tertiary institutions. This research employed a survey design and leaned on the efficient service delivery theory to highlight just how vital quality assurance is for ensuring long-term, high-quality education. Given that quality assurance is key to attaining educational excellence, the data collected from the questionnaires were analyzed using simple percentages and mean scores derived from a four-point Likert scale. The findings revealed that Imo State University in Owerri, Nigeria, falls short in providing quality instruction, largely due to insufficient quality assurance measures. To meet the educational standards set by the Sustainable Development Goals, the study recommended establishing robust quality assurance processes along with effective monitoring and evaluation practices.

Additionally, [Abdullahi \(2020\)](#) looked into the management of the quality assurance process for achieving learning goals in Nigerian secondary schools. The study employed a quantitative research approach. The study's participants included 6,894 instructors and all 310 principals of public senior secondary schools in Kwara State, Nigeria. A sample size of a known population was used to select 175 principals and 364 teachers using a proportional sampling strategy among the three senatorial districts. Using statistical analysis, the *t*-test was used to examine the acquired data. The findings showed that reaching educational goals in Nigeria, quality control, teacher preparation, and adequate funding did not significantly differ from one another. However, there is little research, if any, on managing quality assurance for the sustainability of globalized education systems. Furthermore, no research uses self-evaluation and capacity, and capability as critical variables to measure quality assurance for effective sustainability of the globalized education system. Another significant gap is the use of system theory, which deviates significantly from the preceding research theories. The objective of this study is to close the gaps that earlier researchers left.

The following research objectives can be formulated:

1. To examine the relationship between capacity and capability, and sustainability of globalized education systems
2. To assess the impact of self-evaluation practice on sustainability of globalized education systems.
3. To determine the impact of managing quality assurance on sustainability of globalized education systems

Research questions

1. How do capacity and capability affect the sustainability of globalized education systems?
2. How does self-evaluation practice affect the sustainability of globalized education systems?
3. Does managing quality assurance affect the sustainability of globalized education systems

Research hypotheses

The following hypotheses were created and put to the test:

H1: There is a significant positive relationship between capacity and capability, and sustainability of globalized education systems

H2: There is a significant positive relationship between self-evaluation practice and the sustainability of globalized education systems

H3: There is a significant positive relationship between managing quality assurance and the sustainability of globalized education systems

Review of related literature

Quality assurance (QA) in Nigerian institutions plays a vital role in ensuring that instruction and learning meet high standards. The significance of quality education has grown, particularly as education is recognized as a key driver of economic competitiveness in a knowledge-driven global economy ([Asiyai, 2015](#)). [Ewuzie \(2020\)](#) emphasizes that countries should aim to cultivate a workforce equipped with advanced skills and maintain a competitive edge in research and knowledge production for societal development. The concept of quality is complex and varies among stakeholders. [Okoro \(2015\)](#) defines quality as the fulfillment of widely accepted principles or standards, which may be set by professional organizations, accreditation agencies, or legal frameworks. Essentially, quality distinguishes services or outcomes that meet or exceed recognized standards.

Quality assurance refers to the organized processes, methods, and activities used to ensure that institutional outputs meet or

surpass set standards (Abdullahi, 2025). According to Usman and Madudili (2021), QA involves managing resources effectively, improving processes, and raising output standards to achieve objectives while ensuring public accountability. Its principles, accountability, improvement, and control, are reflected in international QA frameworks such as the UNESCO Guidelines for QA in Higher Education (UNESCO, 2015) and the European Standards and Guidelines for QA [European Association for Quality Assurance in Higher Education [ENQA], 2015]. Accountability ensures stakeholder expectations are met, control maintains resource efficiency, and improvement promotes iterative refinement of processes and outcomes.

QA activities assess whether students achieve their potential and whether institutional resources effectively support learning. The Federal Ministry of Education [Federal Ministry of Education [FME], 2014] stresses that systematic QA ensures evidence-based management of educational standards. Abdullahi (2020) notes that administrators face ongoing challenges in prioritizing and implementing management strategies to achieve high-quality outcomes. QA, therefore, plays a critical role in building institutional credibility and positive reputation. Awodun and Boris (2020) describe QA in academic institutions as the systematic observation, assessment, control, and documentation of practices and programs to maintain accepted standards. Globally, similar principles are observed in quality frameworks such as the Accreditation Board for Engineering and Technology (ABET), (2026) in the United States and the Quality Assurance Agency (QAA) (2026) in the United Kingdom, which emphasize monitoring, evaluation, and continuous improvement as key pillars for sustaining educational quality.

Managing QA involves building capacity, enhancing institutional capabilities, and engaging in self-evaluation. Capacity-building develops knowledge, skills, leadership, and organizational structures needed for effective operation. Wey-Ameawhule and Dikeogu (2023) highlight that capacity-building allows lecturers to enhance subject-matter expertise, innovate pedagogical methods, uphold professional ethics, and engage collaboratively in curriculum and resource management. Internationally, initiatives such as the European Training Foundation programs or UNESCO capacity-building workshops mirror these objectives, underscoring the global emphasis on developing institutional human capital.

Capacity-building is all about developing the knowledge, skills, commitment, structures, systems, and leadership needed for an organization to run effectively. It includes efforts to enhance institutions on three fronts: boosting the knowledge and skills of practitioners, expanding the support and infrastructure for promoting organizations, and fostering collaboration and partnerships. This process allows lecturers to deepen their subject-matter expertise, explore innovative teaching methods, uphold professional ethics, and gain valuable experiences while engaging thoughtfully with colleagues and experts on the theories, methods, practices, and challenges tied to curriculum implementation and resource management in educational settings (Wey-Ameawhule and Dikeogu, 2023). Typically, capacity-building programs are designed for instructors at specific training facilities, whether within or outside of an educational institution, depending on where the participants work, the type of training, and the schedule. The goal is to enhance lecturers' capacity to boost their pedagogical knowledge and skills, ensuring they adopt best practices and

improve efficiency in their teaching methods, ultimately leading to better learning outcomes in schools. The capability strategy can be incredibly useful for analyzing education and schooling, as it provides a framework for assessing educational equality (we should always aim for equality of capabilities) and serves as a strong rationale for education (since it can expand an individual's range of capabilities).

Sustainability involves fostering growth that meets current development goals while ensuring that essential educational systems and resources remain viable for future generations. Anene (2021) defines sustainable development in education as meeting today's learning needs without compromising future generations' opportunities. Globally, sustainable QA practices are embedded in frameworks such as the European Association for Quality Assurance in Higher Education [ENQA] (2015), which emphasize long-term institutional development, responsible resource management, and alignment with international educational standards. In Nigerian universities, sustainability in globalized education entails integrating sustainable practices into curricula, governance, and institutional strategies.

Systems theory provides a useful lens for understanding QA as a dynamic, interdependent system (von Bertalanffy, 1972). Inputs, including teachers, infrastructure, funding, and supervisory personnel, interact with processes such as curriculum delivery and teaching-learning activities to produce outputs, including student achievement, skills acquisition, and overall academic behavior. Feedback loops are critical for monitoring progress and identifying areas for improvement. International QA models, including those promoted by UNESCO and the ESG, similarly emphasize ongoing monitoring, evaluation, and feedback as central to sustainable institutional quality. By focusing on continuous improvement, collaboration, and recognition of students as primary beneficiaries of education, QA systems, such as locally and globally, aim to foster learning environments that are effective, adaptable, and aligned with international standards.

While much of the literature highlights quality assurance within Nigerian higher education, these insights are conceptually and practically linked to global QA discourse. International frameworks, such as UNESCO Guidelines for Quality Assurance (UNESCO, 2015), the European Standards and Guidelines (European Association for Quality Assurance in Higher Education [ENQA], 2015), and accreditation systems like ABET and QAA, share common principles of accountability, continuous improvement, and feedback-driven evaluation, which are mirrored in Nigerian practices. By situating local findings within these broader international perspectives, it becomes evident that effective QA mechanisms in Nigeria reflect global trends, including capacity-building, context-sensitive evaluation, and sustainability-oriented governance. This integration underscores that Nigerian universities, while operating within national contexts, are engaging in processes that parallel and inform global debates on quality assurance and the sustainability of higher education systems.

Theoretical contribution

This study is grounded in the theoretical framework of systems theory, as proposed by von Bertalanffy, 1972. Essentially, systems theory examines how interdependent processes influence each

other over time, contributing to the sustainability of a larger whole. When it comes to education, this theory highlights it as a complex, interconnected system where various elements, like students, teachers, resources, policies, and external factors, interact to achieve desired outcomes. Systems theory helps us grasp how these different components collaborate to uphold and enhance educational standards on a global scale.

This study repositions systems theory, which was first introduced by Ludwig von Bertalanffy, as an analytical framework for comprehending how quality assurance (QA) mechanisms function within globally interconnected higher education systems, going beyond its descriptive application to educational organizations. Systems theory views universities as open, adaptable systems embedded in complex national and international regulatory settings, as opposed to seeing universities as separate institutional units. A more thorough description of how local QA practices are influenced by and react to international standards, norms, and policy flows is made possible by this approach.

Universities operate as open systems with input-throughput-output dynamics, according to systems theory. Regulatory requirements, international benchmarking measurements, financial constraints, technology infrastructures, and labor market expectations are examples of inputs. Peer review, curricular alignment, institutional self-evaluation, and internal governance procedures are all part of throughput. Graduate competences, research productivity, institutional rankings, and international recognition are examples of outputs. Importantly, systems theory's core idea of feedback loops explains how QA results change future institutional behavior, maintaining or adjusting educational quality over time.

Quality assurance (QA) dynamics within globalized education systems can be understood through three interrelated analytical mechanisms identified in this work, which go beyond a descriptive application of systems theory. The first way that QA functions is through cross-scale dependency. Nigerian public universities are subject to national regulations, but universities are also part of a larger global quality architecture that is influenced by transnational ranking systems, international accreditation frameworks, digital learning standards, and cross-border academic partnerships. It is made clearer by systems theory that institutional practices are adaptive reactions to prevailing policy norms and transnational expectations rather than just domestic administrative routines. Therefore, the Nigerian case illustrates how national regulatory and cultural contexts filter, translate, and institutionalize global quality discourses. Therefore, local quality assurance systems mirror both national governance frameworks and international policy currents, exposing the complex and interconnected nature of modern higher education institutions.

Second, without eliminating contextual heterogeneity, globalization creates structural similarities. Even though many higher education systems have come together around similar QA tools, like self-assessment reports, periodic accreditation cycles, and external peer review, differences in resource capacity, governance customs, institutional histories, and organizational culture continue to have a significant impact on how these tools are operationalized. According to the notion of equifinality, which is explained by systems theory, several institutional configurations can pursue and possibly accomplish comparable systemic goals, such as improving quality, even when the institutions

operate under distinct material and regulatory circumstances. By showing that convergence in form does not imply consistency in function or outcome, the Nigerian setting thus offers theoretical insight into how QA systems emerge within circumstances characterized by financial constraint, infrastructure limits, and regulatory centralization.

Third, adaptability rather than strict compliance processes is more important for the viability of worldwide educational institutions. QA procedures typically encourage procedural compliance without improving systemic responsiveness when the procedures serve largely as bureaucratic reporting exercises. Nonetheless, when they are created as reflexive feedback systems, they help organizations to reassess procedures, boost output, and foster organizational learning. Depending on how feedback loops are read and implemented, internal self-evaluation can either strengthen compliance-oriented routines or foster cultures of continual improvement, as demonstrated by data from public colleges in Nigeria. This contrast adds to broader theoretical discussions about the transition from accountability-driven QA models to enhancement-oriented strategies that put long-term sustainability and institutional resilience first in globally interconnected higher education institutions.

Despite the empirical focus on public universities in Nigeria, the study does not seek to statistically generalize across global systems. Rather, it provides analytical generalization: the Nigerian case provides a conceptually illuminating site for investigating the internalization, negotiation, and operationalization of global QA norms in a public higher education system with limited resources. Institutions outside the dominant Western contexts frequently have to balance local socioeconomic realities with expectations to meet global benchmarking standards in globally stratified educational systems. The study advances knowledge of how peripheral or semi-peripheral systems maintain legitimacy in a global higher education field without completely reproducing high-resource institutional models by using systems theory to theorize this dynamic. As a result, the Nigerian case serves as an example of a larger class of systems functioning under comparable structural circumstances throughout Africa, parts of Asia, and Latin America.

This study advances systems theory as a framework for comprehending quality assurance (QA) as a dynamic and multi-scalar governance mechanism functioning across interconnected educational environments, rather than just as an administrative tool. This theoretical contribution to global higher education scholarship is significant. The paper illustrates how constant feedback loops between globally circulating norms and their localized implementation impact institutional practices by framing QA within an open-systems framework. These repeated adaptation processes show that quality assurance is shaped by the interplay between national institutional realities and international expectations, rather than being totally context-bound or standardized.

The study also presents a context-sensitive analytical model that clarifies how differences in funding, infrastructure, and regulatory customs affect how QA systems are set up and function without necessarily undermining their fundamental quality goals. By doing this, it disproves the notion that successful QA needs to imitate institutional models with many resources and instead emphasizes the flexible ways in which institutions work to improve quality in a variety of

structural conditions. Lastly, the paradigm established in this research offers a foundation for cross-national comparative analysis within uneven global higher education hierarchies. This study moves beyond a descriptive description of Nigerian public universities and presents the case as philosophically instructive by analytically elucidating how institutions respond to global influences while negotiating local limits. It contributes to more general discussions on governance, quality, and sustainability in modern higher education by showing how adaptive, interdependent institutional mechanisms support international educational institutions.

Methodology

Research design

This research employs a mixed-approach strategy, combining quantitative and qualitative methods to improve validity via methodological triangulation. While the quantitative component employs standardized surveys to examine statistical correlations between capacity and capability, self-evaluation, and sustainability, the qualitative component provides deeper insights into stakeholder viewpoints and contextual practices. This triangulation enhances the findings' credibility by confirming data sources and methodologies (Creswell, 2015).

Population and sampling procedure

The study was conducted across three purposively selected universities in Nigeria: one Federal university, one State university, and one Private university. The selection process was guided by theoretical considerations rather than mere convenience. Nigeria's higher education landscape is quite diverse, with federal, state, and private institutions each following their own funding models, regulatory challenges, administrative styles, and resource availability. By including one university from each type of ownership, the study aimed to reflect the systemic differences present in the national higher education framework. This diversity among institutions played a crucial role in shaping the study's systems-theoretical framework. Systems theory highlights the importance of interdependence, adaptability, and the need to differentiate contextually within complex organizational settings. Federal universities mainly receive funding and oversight from the national government, often functioning within larger, more bureaucratic systems. On the other hand, State universities, while also publicly regulated, deal with different financial challenges and political scrutiny at the State level. Private universities tend to depend heavily on tuition fees, which gives more managerial flexibility, but the institutions still have to meet strict accreditation standards. By looking at these three types of institutions, the study aimed to explore how quality assurance (QA) mechanisms are organized, implemented, and maintained under different governance and resource conditions. The goal wasn't to statistically represent all Nigerian universities, but rather to provide analytical insights across key institutional types within the national framework.

The study focused on three key groups involved in quality assurance (QA) processes: lecturers, students, and QA personnel. To gather quantitative data, researchers used a stratified sampling method, ensuring that each group was fairly represented across different faculties and departments in the chosen universities. Out of 500 questionnaires distributed, 400 valid responses were analyzed. This stratification helped maintain the study's internal validity by preventing any one group or academic discipline from skewing the results. For the qualitative part of the study, 15 in-depth interviews were conducted with a targeted sampling approach to gain insightful views on QA practices and sustainability. The participants included five lecturers, five QA officials, and five senior students who had firsthand experience with accreditation processes, internal quality reviews, or self-evaluation. Including stakeholders at multiple levels enriched the study's depth and allowed for a well-rounded analysis of governance, implementation, and the perspectives of those affected within various institutional frameworks.

Instrumentation

This study employed a mixed-methods approach, utilizing structured questionnaires, semi-structured interviews, and document analysis. Greater methodological transparency was provided to shed light on the instrument adaptation process, operationalized constructs, and integrated strategies into the design. The structured questionnaire was distributed electronically through Google Forms. It focused on three key constructs grounded in theory: (1) capacity and capability, (2) self-evaluation practices, and (3) sustainability in globalized education systems. The instruments were adapted from previously validated instruments used in higher education quality assurance and research on institutional sustainability. This adaptation process unfolded in three stages. First, the researchers conducted a systematic review of quality assurance and sustainability measurement tools to pinpoint scales that had established reliability and conceptual alignment. Next, we reworded the items to better fit the specific regulatory context of Nigerian higher education while keeping their original intent intact. Finally, researchers cross-checked the alignment of constructs against the study's systems-theoretical framework to ensure that each subscale represented distinct yet interconnected dimensions of how institutions function. To establish content validity, two experts in educational management and quality assurance reviewed the items, assessing them for clarity, relevance, and redundancy. Their feedback led to minor wording adjustments and the removal of overlapping items, enhancing the distinctiveness of each construct. Researchers also conducted a pilot study with 30 academic staff members to test the reliability and clarity of the items. The internal consistency coefficients were impressive: capacity and capability ($\alpha = 0.87$), self-evaluation ($\alpha = 0.85$), and sustainability ($\alpha = 0.89$). These figures surpass the conventional threshold of 0.70, indicating that our scales are reliable. All items were rated on a five-point scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

A sequential explanatory mixed-methods design was used for the investigation. First, quantitative data were gathered and examined to find trends in sustainability attitudes, self-evaluation techniques, and capability. Also, the qualitative phase

was conducted to clarify, elaborate, and put the quantitative results in context. Insight into the systemic mechanisms behind survey replies, statistical patterns, and surprising outcomes was all clarified using interview data.

Concurrent with the qualitative phase, a document analysis of institutional QA policy manuals was carried out in order to verify statements regarding formal policy structures versus practices that were reported. Integration took place during the interpretation phase, when systems theory was used to synthesize quantitative trends, interview narratives, and documentary evidence.

The constructs were operationalized to improve conceptual clarity: Capacity and capability are the organizational competencies and institutional resources required to maintain quality assurance procedures. Financial resources, employee knowledge, professional development programs, digital infrastructure, and administrative support systems are all included in this. The “input” dimension of a systems framework is reflected in it. Self-evaluation is the term used to describe organized internal monitoring and assessment procedures that produce input for institutional development. This comprises cycles of curriculum reviews, internal audits, stakeholder input systems, and recurring academic reviews. It stands for the “throughput” or feedback-loop element of how well a system works. Sustainability in a Globalized Education System was defined as the ability of the institution to sustain educational quality over time while conforming to changing global benchmarking demands, cross-border academic norms, digital transformation trends, and international standards. This concept employed quantifiable indicators, such as conformity to international accreditation standards, responsiveness to global labor market expectations, participation in international partnerships, and adaptive policy frameworks, instead of merely referring to globalization in an abstract manner. From an open-systems approach, it represents the “output” and adaptive sustainability dimensions.

Data collection technique

This study took a comprehensive approach to data collection by blending both quantitative and qualitative methods, which helped to provide a well-rounded view of quality assurance (QA) practices. For the quantitative part, we used a structured 25-item questionnaire that was distributed online through Google Forms. We gathered responses from 400 academic staff and students at public universities in Nigeria. The survey aimed to understand perceptions regarding institutional capacity, capability, self-evaluation, and sustainability, and we analyzed the results using SmartPLS 3.0. To enhance the validity of our findings and address the limitations of relying on a single method, we also collected qualitative data through semi-structured interviews and document analysis. We conducted fifteen interviews with key stakeholders, including five lecturers, five QA officers, and five senior students. The questions revolved around essential QA themes like the alignment of policy and practice, as well as stakeholder engagement. We transcribed the interviews and analyzed them thematically. Additionally, we reviewed QA policy documents from three universities to assess how well institutional practices aligned with formal QA frameworks. This combination of methods, interviews, and document reviews allowed for a richer interpretation and verification of the data. It ensured that our

findings were not only statistically sound but also contextually rich, adding credibility and depth to the study.

Data analysis

The study employed a mixed-methods approach to data analysis, which means it combined both quantitative measurements and qualitative insights. This blend provided a richer understanding of how quality assurance (QA) mechanisms operate in the realm of globalized education. For the quantitative side, the analysis utilized SmartPLS 3.0, a structural equation modeling (SEM) tool that's great for working with latent constructs and smaller sample sizes. The analysis unfolded in two main stages: first, the measurement model was evaluated to check the reliability and validity of the constructs, using benchmarks like Cronbach's alpha, composite reliability, and average variance extracted (AVE). Next, the structural model was scrutinized to explore the proposed relationships between capacity and capability, self-evaluation, and sustainability (Hair et al., 2020). On the qualitative front, data from semi-structured interviews were examined through thematic analysis, following Braun and Clarke's six-step model. The transcribed interviews were coded, and patterns were organized into significant themes such as capacity gaps, stakeholder engagement, and the disconnect between policy and practice. This thematic framework not only helped interpret the statistical findings but also offered a deeper insight into institutional behaviors.

Findings and discussion

Measurement model assessment

The validity and accuracy of a measurement model are important to assess properly. One key aspect is the composite reliability measure, which should ideally be above 0.7 to ensure the construct's reliability. Similarly, to confirm an item's reliability, we need a score greater than 0.70 along with the evaluation of the outer loadings measure (Hair et al., 2017). Additionally, the Average Variance Extracted (AVE) should exceed 0.5. The results were summarized in Figure 1 and Table 1.

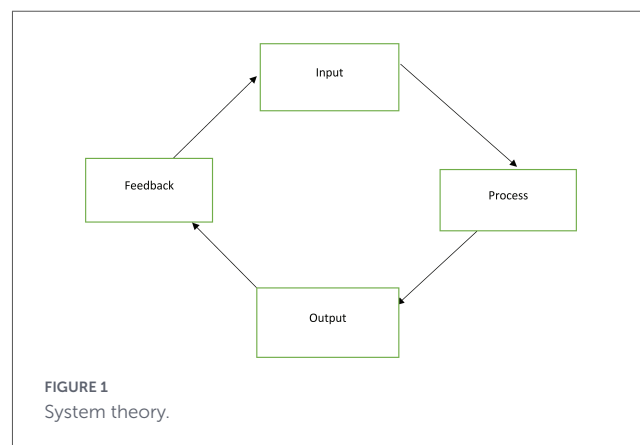


TABLE 1 Summary of measurement model (construct reliability and validity) ($p < 0.01$).

Constructs	Items	Factor loading	Cronbach's alpha	Composite reliability	Average variance extracted
Capacity and capability (CC1)	The financial resources of my organization are adequate to support quality assurance initiatives	0.721	0.861	0.891	0.511
CC2	There is enough personnel devoted to academic monitoring and quality control.	0.598			
CC3	Regular professional development about accrediting standards and teaching quality is provided to faculty and staff.	0.503			
CC4	The organization has sufficient digital infrastructure to facilitate quality monitoring, such as data systems and an LMS.	0.731			
CC5	The administrative assistance required to carry out QA policies is given to academic departments.	0.729			
CC6	Leadership exhibits a strong dedication to upholding academic standards.	0.855			
CC7	Quality assurance duties and responsibilities are spelled out in institutional policies.	0.782			
CC8	The technological know-how needed to match national and international quality standards with its programs is present at my institution.	0.741			
Self-evaluation (SE1)	My institution regularly conducts structured internal academic reviews.	0.867	0.864	0.894	0.516
SE2	Departments use performance data (e.g., student results, completion rates) to inform improvement decisions.	0.690			
SE3	Student feedback is systematically collected and integrated into program evaluation.	0.678			
SE4	Internal quality audits are conducted before external accreditation visits.	0.781			
SE5	There are clear procedures for curriculum review and revision.	0.750			
SE6	Self-assessment reports are prepared and discussed transparently within the institution.	0.635			
SE7	Findings from internal evaluations lead to measurable improvements	0.527			
SE8	Academic staff are actively involved in institutional self-evaluation processes.	0.735			
Sustainable globalized education (SGE1)	My institution's quality assurance practices are aligned with recognized international standards.	0.767	0.911	0.927	0.587
SGE2.	Academic programs are periodically reviewed to reflect global developments in the discipline.	0.800			
SGE3.	My institution benchmarks its performance against international universities.	0.869			
SGE4	Quality assurance processes consider global labor market expectations.	0.855			
SGE5.	My institution participates in international academic collaborations.	0.786			
SGE6.	My institution demonstrates the capacity to sustain educational quality during periods of policy.	0.729			
SGE7	Policies are adaptable to emerging global trends in higher education (e.g., digitalization, internationalization).	0.855			
SGE8	QA systems support long-term institutional strategic planning	0.670			
SGE9	Quality assurance practices enhance the institution's global competitiveness and reputation.	0.685			

To determine construct reliability, we used Cronbach's Alpha and Composite Reliability (CR). Table 1 provides a clear view of the convergent validity and construct reliability for each construct. For all constructs, the Cronbach alpha values and CR were above the recommended threshold of 0.700 across the entire sample. This supports the convergent validity, as the Average Variance Extracted (AVE) for the constructs was greater than 0.500 for the whole sample (Diamantopoulos et al., 2012).

The measurement model results in Table 1 indicate that all three constructs: Capacity and Capability (CC), Self-Evaluation (SE), and Sustainable Globalized Education (SGE), demonstrate adequate reliability and validity. For Capacity and Capability, factor loadings ranged from 0.503 to 0.855, with a Cronbach's alpha of 0.861, composite reliability of 0.891, and an AVE of 0.511, indicating that the items consistently measure the intended construct. Self-Evaluation showed similarly strong properties, with loadings from 0.527 to 0.867, Cronbach's alpha of 0.864, composite reliability of 0.894, and AVE of 0.516, confirming internal consistency and convergent validity. The Sustainable Globalized Education construct exhibited the highest reliability, with loadings between 0.670 and 0.869, Cronbach's alpha of 0.911, composite reliability of 0.927, and AVE of 0.587, reflecting strong convergent validity and consistent measurement. Thus, the results demonstrate that the survey items are reliable indicators of their

respective constructs, supporting their use in subsequent structural analysis.

Figure 2 presents the structural model showing the relationships between Capacity and Capability (CC), Self-Evaluation (SE), and Sustainable Globalized Education (SGE). The figure indicates that both Capacity and Capability and Self-Evaluation positively influence SGE, with path coefficients of 0.458 and 0.395, respectively. All measurement items load significantly on their respective constructs, with factor loadings ranging from 0.503 to 0.869, demonstrating that the indicators reliably reflect their latent variables. The SGE construct has a strong R^2 value of 0.636, suggesting that approximately 63.6% of the variance in sustainable globalized education is explained collectively by institutional capacity, capability, and self-evaluation practices. Therefore, the model confirms that both human and institutional resources, along with structured internal evaluation processes, are critical predictors of sustainable quality assurance aligned with global standards.

Additionally, the Fornell and Lacker criteria were used to assess the discriminant validity of the construct. HTMT was also evaluated to evaluate in-depth discriminant validity. The values in the diagonals should lie under the associated values in this technique, and they should be higher than them. The diagonals show the AVEs' square root, which ought to be greater than the correlation between their variables. The diagonal values in Table 2 are greater than their corresponding values, which satisfies the

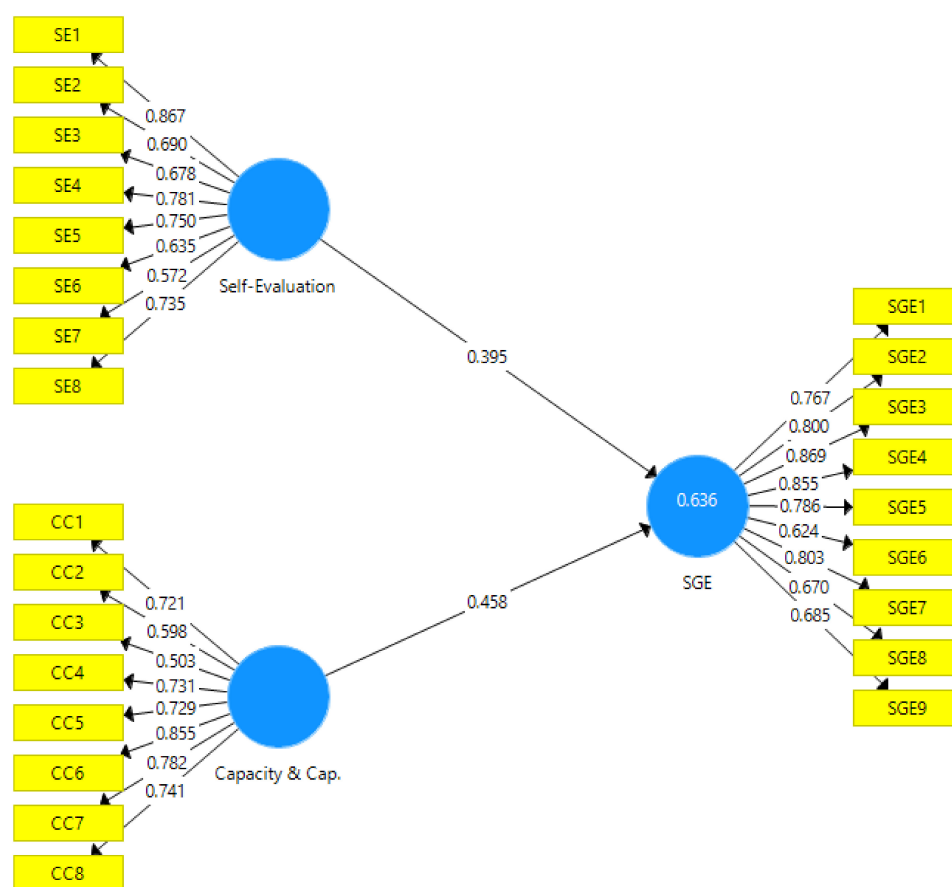


FIGURE 2
Measurement model of the constructs.

TABLE 2 Fornell and Lacker criterion.

	CC	SE	SGE
CC	0.715	0.719	0.766
SE	0.748		
SGE	0.753	0.766	

AVE's square root is the diagonal, and the correlations between latent variables are represented by the off-diagonal values.

Fornell and Lacker criterion and upholds discriminant validity (Flake et al., 2022).

Moreover, the discriminant validity of the construct was evaluated utilizing the HTMT. Since the HTMT criteria compute the geometric mean of the average correlations between indicators measuring the same construct and the mean of all correlations between indicators measuring different constructs, they are regarded as reliable for determining discriminant validity (Flake et al., 2022). Table 3 displays the HTMT values. If this measure's values are less than 0.85, it indicates that the variables are not the same. Also, it can be concluded that this data has no issue of discriminant and convergent validity, showing that the data has been collected fine because HTMT is not greater than 0.85. HTMT is the latest criterion for testing discriminant validity, and it is based on internal correlation and external correlation.

Assessment of the structural model

The structural model is depicted visually. Figure 3 depicts the graph, and the arrows linking the study's constructs were decided by the direction of the hypotheses presented in the framework. The single-headed arrows are used to confirm the significance of the study concept. Figure 3 illustrates the factor loadings for each item.

Hypothesis analysis

The results for the hypothesis have been given in Figures 3, 4 and Table 3. It shows that self-evaluation has an effect of 0.393 on the Sustainability of globalized education systems with a p -value of 0.000, which is less than 0.05 and a t -value greater than 1.96, which means there is a relationship between ES and SGE. Also, A relationship between CC and SGE is evident as SE has an effect of 0.460 on SGE with a p -value of 0.000, which is less than 0.05, and a t -value larger than 1.96. According to the results, the hypothesis has been accepted, as the β has a value greater than the criterion values (0.10), which means that all the constructs have significant contributions and impact on the globalized education systems. In the same way, the t -statistics for the constructs have a greater value than the recommended one, i.e., (1.96), which indicates the logical dependence of the dependent variable on the independent variables. In summary,

TABLE 3 HTMT Values of the construct.

	CC	SGE	SE
CC	0.798	0.803	
SGE			
SE	0.856		

the results show that the assertiveness of the hypotheses was supported.

Figure 3 presents the structural model with significance testing of the relationships between Capacity and Capability (CC), Self-Evaluation (SE), and Sustainable Globalized Education (SGE). The model shows that both Capacity and Capability and Self-Evaluation have a statistically significant positive effect on SGE, with t -values of 0.000 for $CC \rightarrow SGE$ and $SE \rightarrow SGE$, indicating strong significance at conventional levels. The item-level t -values range widely, demonstrating that each indicator contributes meaningfully to its respective construct. Thus, the figure confirms that institutional resources, capabilities, and internal evaluation processes are critical determinants of sustainable quality assurance in globalized educational systems, and the paths are statistically significant.

Figure 4 illustrates the structural relationships between Self-Evaluation, Capacity and Capability, and Sustainable Globalized Education (SGE). Both Self-Evaluation ($\beta = 0.393$, $p < 0.001$) and Capacity and Capability ($\beta = 0.460$, $p < 0.001$) have positive and significant effects on SGE, indicating that higher stakeholder competence and reflective self-assessment are associated with improved SGE outcomes. The figure also shows the individual observed indicators (SE1–SE8 for Self-Evaluation, CC1–CC8 for Capacity & Capability, and SGE1–SGE9 reviewed reviewed for SGE) and their respective factor loadings, demonstrating that all items contribute meaningfully to their latent constructs.

Table 4 shows that both Capacity and Capability ($\beta = 0.460$, $t = 4.991$, $p < 0.001$) and Self-Evaluation ($\beta = 0.393$, $t = 4.073$, $p < 0.001$) have positive and significant effects on Sustainable Globalized Education (SGE). Both hypotheses are supported, suggesting that enhancing stakeholder competence and promoting self-assessment contribute significantly to effective SGE outcomes. Hence, the hypothesis (H2) was supported.

Qualitative findings and discussion

To better understand how quality assurance procedures impact institutional performance and sustainability in globalized educational systems, two visual models, a flowchart and a mindmap, were created. The logical paths of impact and the structural themes are both depicted in the diagrams. Here is the presentation of the qualitative data. The analyzed data were collected through interview questions, which were analyzed with the use of thematic analysis in the following order in Figures 5, 6.

The mindmap paints a detailed picture of the complex web of themes that influence the sustainability of education systems in a globalized world. At the heart of the diagram is the main goal: Sustainability of Globalized Education, which highlights the importance of academic institutions being viable, adaptable, and relevant on a global scale. From this central concept, five key thematic areas emerge from the qualitative analysis conducted in the study:

Research Question 1: What are the primary challenges your institution encounters regarding staff training, funding, or strategic planning for quality assurance?

Theme 1: Capacity and Capability Gaps Undermine QA Systems

Theme 1: QA Systems Are Affected by Capacity and Capability Gaps

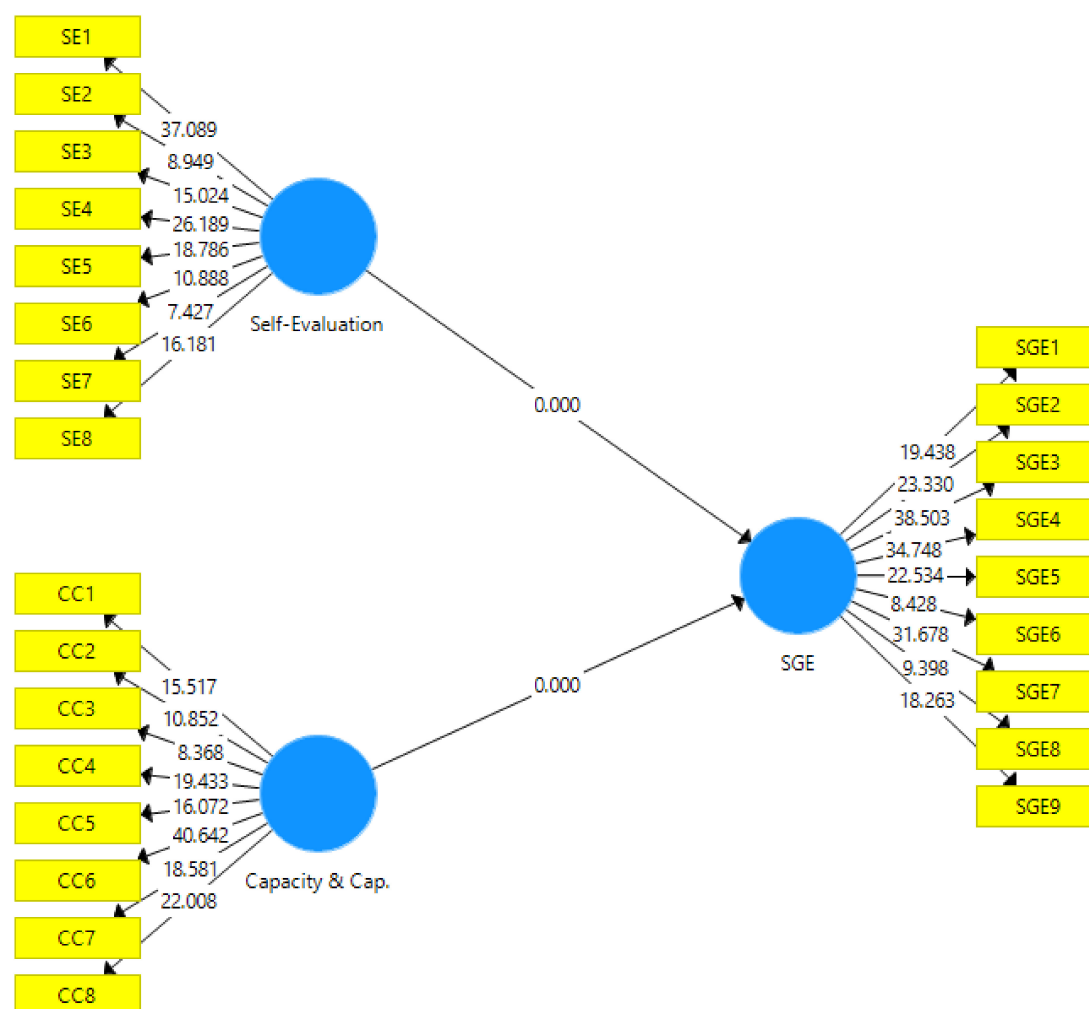


FIGURE 3
Structural Model of the Study.

Effective QA implementation was frequently observed to be hampered by capacity issues, which include limited finance, inadequate personnel training, and poor strategic planning. Such resource shortages make it more difficult to operationalize QA frameworks, according to [Abdullahi \(2018\)](#). According to systems theory, these are deficiencies in system inputs that compromise outputs and lessen feedback loop effectiveness ([von Bertalanffy, 1972](#)). Quantitative results highlight the crucial importance of institutional capability by confirming that organizations with higher capacity ratings exhibit superior sustainability outcomes.

Research Question 2: How does your institution conduct self-evaluation processes, and do they result in any significant changes or improvements?

Theme 2: Self-Evaluation as a Missed Opportunity

Internal review procedures frequently served more as formalities than as tools for significant progress. This supports the findings of [Ewuzie \(2020\)](#) and [Anene \(2021\)](#), who point out that self-evaluation only improves responsiveness when it is included in decision-making. Weak self-evaluation lowers throughput efficiency from a systems perspective, which limits the university's ability to adjust even with QA procedures in place.

Research Question 3: Based on your experience, how effectively are your institution's quality assurance (QA) policies put into practice in everyday academic or administrative activities?

Theme 3: The Gap Between QA Policies and Actual Practice

There is a gap between intended and actual behaviors since many organizations save manuals and strategic documents, but they are frequently underutilized. In situations with limited resources, [Wey-Ameawhule and Dikeogu \(2023\)](#) draw attention to such policy-practice discrepancies. This is framed by systems theory as decoupling: formal structures exist, but systemic inefficiencies prevent the intended outputs from being realized.

Research Question 4: How involved are faculty, students, and administrative staff in shaping or reviewing the quality assurance practices at your institution?

Theme 4: The Role of Stakeholder Engagement in QA Success

QA is more legitimate and responsive when teachers, students, and administrative staff participate inclusively. Stakeholder participation is also emphasized by international frameworks like the European Standards and Guidelines (ESG). According to the study, participation improves feedback loops, which enhances system cohesiveness and the precision of self-evaluation findings.

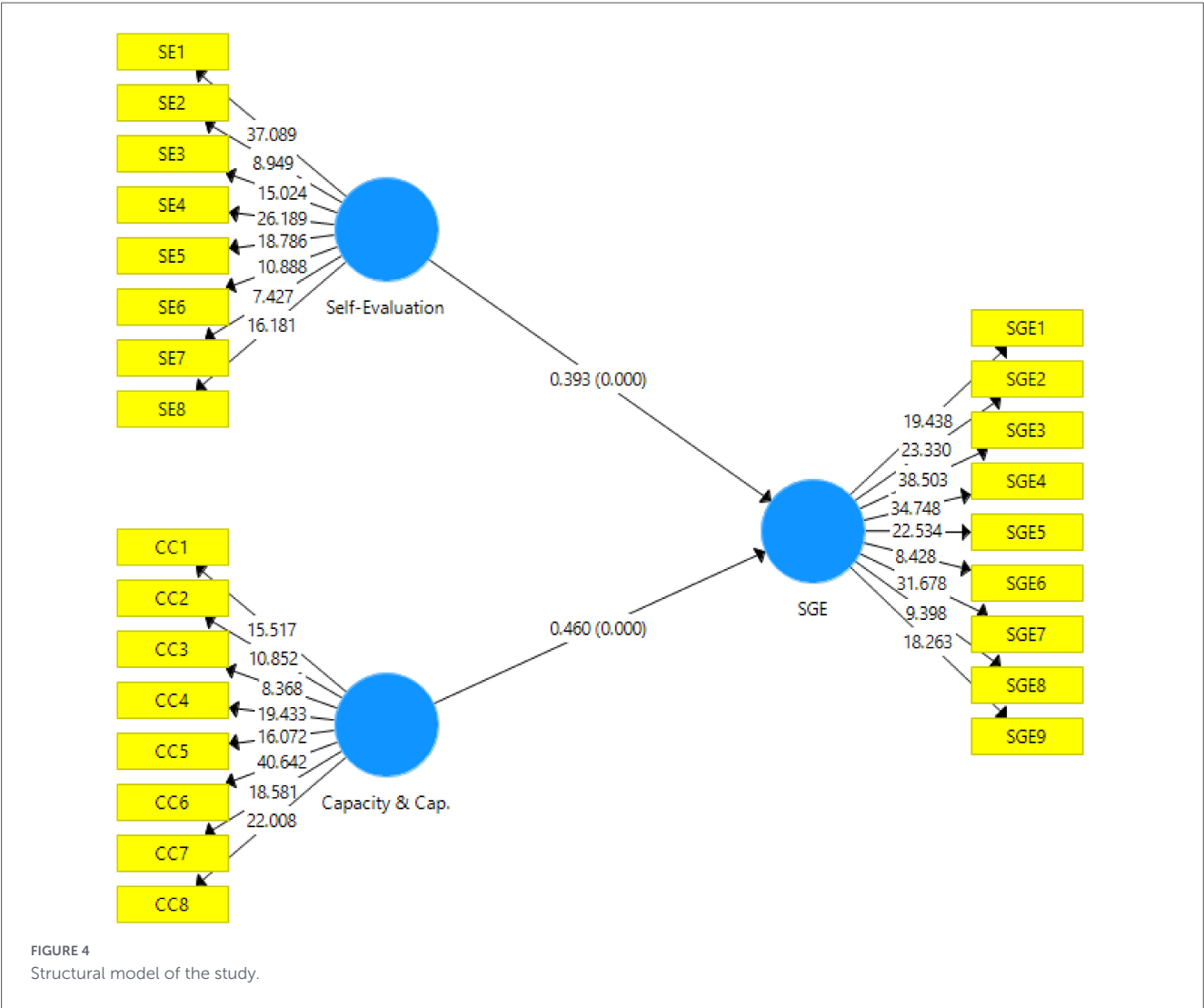


TABLE 4 Summary of path coefficients.

Hypotheses	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics	P-Value	Decision
Capacity and Cap. - > SGE	0.460	0.467	0.092	4.991	0.000	Support
Self-Evaluation - > SGE	0.393	0.385	0.097	4.073	0.000	Support

Research Question 5: Do you believe that the QA strategies at your institution truly reflect local realities and challenges, or do they seem imposed and disconnected?

Theme 5: The Need for Contextual QA

Lastly, QA tactics need to be adjusted to local conditions. If standardized models are used without being adjusted for institutional circumstances, cultural norms, or regulatory settings, they may not be effective. According to Anene (2021) and Abdullahi (2020), context-sensitive QA improves sustainability and relevance. The mindmap emphasizes how capacity, evaluation, stakeholder engagement, and contextual adaptation interact dynamically to produce effective QA.

According to the flowchart (Figure 6), QA serves as the primary intermediary between institutional resources, procedures,

and results. Sustainability and global competitiveness are two important results that are influenced by the dynamic interaction of the five thematic branches: capacity, self-evaluation, policy-practice alignment, stakeholder engagement, and contextualization.

Quantitative analysis confirmed statistically significant relationships between capacity, self-evaluation practices, and sustainability outcomes. Higher institutional capacity and rigorous self-evaluation corresponded with improved resilience, adaptability, and academic performance. The qualitative data provide explanatory depth: capacity gaps, superficial evaluations, and policy-practice misalignment account for variability in QA effectiveness, highlighting the mechanisms underlying the quantitative associations. The results are consistent with other research that highlights QA as a strategic way to improve



FIGURE 5

A mindmap diagram illustrating the interconnected themes that impact the “sustainability of globalized education systems.”

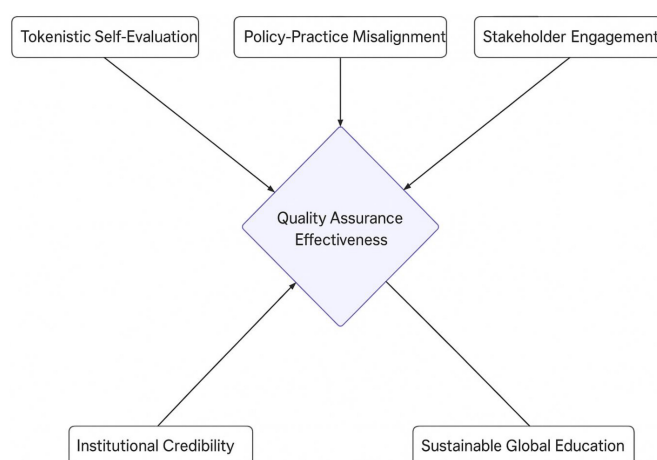


FIGURE 6

Flowchart diagram illustrating the effectiveness of quality assurance.

institutional excellence (Abdullahi, 2018; Ewuzie, 2020; Anene, 2021; Wey-Ameawhule and Dikeogu, 2023), which also addresses existing literature that highlights some of its possible drawbacks. Critiques of global QA systems have expressed concern that audit-driven QA may promote compliance rather than real progress. The study shows that institutional commitment, contextual adaptation, and active feedback loops are crucial due to the fact that formal procedures alone are not enough.

This study operationalizes Ludwig von Bertalanffy's (1972) systems theory by projecting quality assurance (QA) procedures onto an input-throughput-output model of institutional operation. In this approach, the inputs of the system are represented by capacity and capability. These comprise the administrative support systems, staff knowledge, technology infrastructure, and financial resources that offer the fundamental circumstances required for successful QA implementation. The system lacks the stability needed to maintain constant standards or pursue long-term progress in the absence of sufficient inputs. The system's throughput component is made up of stakeholder involvement and self-evaluation procedures. These procedures serve as feedback systems that let organizations keep an eye on performance, incorporate assessment information into decisions, and adjust plans of action in response to demands from both inside and

outside the organization. By guaranteeing that a variety of institutional actors participate in the interpretation and application of evaluation results, participatory engagement enhances these feedback loops.

Sustainability is the system's output and ability to adjust. At this point, the results of successful systemic interaction become evident in the form of institutional resilience, global competitiveness, and the capacity to conform to international quality assurance norms. Crucially, sustainability in international higher education is not a fixed goal but rather a sign of the system's ability to adjust to shifting environmental demands. The results show that flaws in inputs or interruptions in feedback processes severely limit outputs. Put differently, institutional adaptation is weakened by insufficient resources or flimsy self-evaluation approaches. Consequently, the efficacy of quality assurance arises not only from the presence of formal policies but also from the dynamic interplay between resources, procedures, and adaptive results within the institutional

Nigerian universities use comparable procedural structures to international QA models (such as ESG and North American accreditation standards), but universities operate differently because of resource unpredictability, regulatory centralization, and contextual variety. The study demonstrates that, despite the emphasis on standardization in global frameworks, relevance and

efficacy depend on adaptability to local situations. By bridging the gap between formal compliance and functional sustainability, contextualized QA validates Abdullahi's (2020) assertion that culturally and institutionally sensitive QA design is crucial.

The study contributes to the understanding of quality assurance in globalized education by showing that institutional capacity, evaluative rigor, stakeholder involvement, and contextual adaptability must all work together in concert for sustainability. Self-evaluation becomes a crucial leverage point, but only when it is incorporated into effective feedback loops and has sufficient backing. Therefore, QA functions as a dynamic governance mechanism that mediates institutional performance in intricately linked learning contexts. The results critically extend and support previous research, demonstrating that successful quality assurance in international situations is dependent rather than automatic. Systems theory offers a logical framework for comprehending these dynamics, demonstrating that adaptive processes, outputs that are sensitive to local and global stresses, and well-aligned inputs all contribute to sustainability.

Implications and limitations of the study

The study emphasizes the need for governments and higher education institutions to enhance sustainable, globalized education through improved organizational capacity and self-evaluation. A key implication is the integration of sustainability components into the curriculum, focusing not only on theory but also on practical teaching methods. The government should consider professional development initiatives for lecturers that enhance their ability to teach sustainability. The study extends system theory, highlighting the importance of factors influencing self-evaluation and capacity for sustainability. Future research should explore how contextual factors impact the relationship between sustainability and education, using longitudinal studies to track these dynamics over time. Limitations of this research suggest that future studies should investigate these dynamics in greater detail, utilizing comparative and longitudinal approaches.

Conclusion

This study examined how the sustainability of globalized educational systems is influenced by quality assurance mechanisms, specifically institutional capacity and capability, and self-evaluation practices. Using a triangulated mixed-methods approach, combining quantitative surveys, qualitative interviews, and document analysis, the research provides a nuanced understanding of how internal processes, feedback mechanisms, and institutional resources shape long-term educational outcomes. The findings indicate that context-sensitive implementation, institutional adaptability, and investment in human capacity are more decisive for sustainability than the mere presence of formal QA frameworks. Systems theory proves useful in interpreting these results, highlighting the interdependencies among institutional inputs, processes, and adaptive outcomes. In particular, human

capital, academic staff expertise, and engagement emerge as critical drivers of sustained quality improvement. This study suggests that effective QA mechanisms contribute to enhancing the institution's global relevance and competitiveness. It acknowledges that global competitiveness in this study is conceptualized as the institution's ability to align with international quality standards, respond to global educational trends, and maintain institutional resilience in an interconnected academic environment. This conceptual framing is supported by evidence of benchmarking practices, adherence to international QA guidelines, and adaptive responses to globalized educational demands observed in the participating universities, rather than through direct measurement of international rankings or cross-border performance metrics. Therefore, this study advances knowledge by understanding QA as a dynamic, adaptive system capable of enhancing stakeholder confidence, institutional credibility, and global relevance, emphasizing that sustainability depends on active management of both human and institutional capacities rather than compliance alone.

Recommendations

A standardized, strategic approach to quality assurance (QA) is essential for sustaining international education amid global academic integration. Institutions should first enhance QA systems by investing in modern infrastructure, data tools, and skilled personnel. Continuous professional development must be prioritized to align staff with evolving national and international QA standards. Collaboration with global institutions can foster shared best practices, benchmarking, and joint evaluations. Regular, data-driven self-assessments should be embedded into institutional culture, encouraging departments and faculty to reflect on and improve their practices. Digital tools can facilitate dynamic feedback from staff and students, supporting continuous improvement. Strong governance and coherent QA policies are also critical. Governments and regulatory bodies must craft supportive frameworks that allow institutions both guidance and autonomy. QA should be fully integrated into institutional strategic planning and should not be treated as an add-on, so it becomes a driver of growth, innovation, and long-term sustainability.

Data availability statement

The original contributions presented in the study are included in the article/[supplementary material](#), further inquiries can be directed to the corresponding authors.

Ethics statement

The studies involving humans were approved by University of Ilorin, University Ethical Review Committee. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

NA: Writing – original draft, Writing – review & editing, Investigation. BM: Methodology, Writing – review & editing. MM: Investigation, Writing – review & editing, Data curation.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. ChatGPT and Grammarly was utilized

in this study to enhance the quality of written content. Grammarly's recommendations and edits were used to maximize readability and linguistic fluency by improving sentence structure and making some study interpretations more accurate and cohesive.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1810032/full#supplementary-material>

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